

EXTRA_I: correcting and creating meta-analyses with local language models — a pre-registered five-phase benchmark and a six-model, three-anchor replication, under a perturbation reading proof and a self-correcting answer key

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September 2026

Abstract

Language models are entering systematic-review workflows, but most evaluations use cloud models, cannot separate reading from training-data recall, and treat the human answer key as infallible. EXTRA_I is a pre-registered benchmark of quantized open-weight models (8–27B parameters, one consumer mini-PC, integrated GPU, no cloud calls) as meta-analysis workers, under two controls: a *perturbation reading proof* (load-bearing numbers displaced under sealed maps, so that recitation is detectable) and a *two-layer, source-verified answer key* that grades the published review on the same ruler as the models. A five-phase chain (read, calculate, create, orchestrate, deploy) ran five models over two published meta-analyses and their 21 trials; a second campaign, registered before its first call, ran six models as peers over three anchors and under both extraction sheets (696 sheets, 39.7 h of machine time, no refusals). Reading: no attributable recitation in either campaign; the second campaign’s 633 non-matching cells were adjudicated with the source quotation — 288 faithful to the source in another encoding or population layer, 175 omissions, 149 reading slips (16 with values the text does not print) and 21 grader-side artifacts. On the same ruler the published reviews agreed with their sources in 102 of 124, 49 of 49 and 29 of 32 cells. The third anchor’s published estimate (OR 0.73 [0.40; 1.36]) can be reproduced only by reproducing two source-confirmed transcription errors; no model reproduced either, and two 14B models scored 32 of 32 cells and pooled to the corrected estimate, 0.48 [0.34; 0.69], a significant benefit where the review reported none. Arithmetic: 1 of 73 unaided confidence intervals was exact, 86% with a text-protocol calculator. Creation: every model reproduced the dichotomous pools; the continuous estimate separated them. Deployment (clean texts, no answer key): the model sided with the sources at all seven discrepancies reaching a single cell. The mandatory-quotation sheet cost a mean 6.5 cells per model on the descriptive anchor and improved no estimate; a context change of 8,192 tokens moved one control by one cell and is declared a confounder. All protocols, seals, sheets, adjudications and raw outputs are public.

1 Introduction

Systematic reviews and meta-analyses are the reference standard of clinical evidence synthesis, with mature reporting standards [Page et al., 2021], and they are labor-bound: registered reviews take 67 weeks on average from registration to publication [Borah et al., 2017], with extraction, checking and pooling consuming most of a team’s hours. Large language models are entering exactly this workflow: feasibility studies report promising but uneven accuracy for

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trial data extraction [Schmidt et al., 2024, Khraisha et al., 2024, Konet et al., 2024]; assisted extraction proved noninferior to human-only extraction in a live review [Lee et al., 2026]; screening assistance is approaching routine use [Xie et al., 2026]; and end-to-end platforms now report complete reviews in hours [Bakker et al., 2026], with fully automated drafts rated above a published human-authored review by blinded specialists [McLaughlin et al., 2026]. The major evidence-synthesis organizations jointly condition such use on demonstrated rigor, human oversight and transparent reporting [Fleming et al., 2025], and a reporting guideline for studies that use large language models now exists [Gallifant et al., 2025].

Three gaps run through this literature. First, most evaluations use commercial cloud models, while clinical text pushes the other way: local open-weight models keep sensitive material on-premises and have shown extraction performance approaching expert annotation [Wiest et al., 2024, Spaanderman et al., 2025]. Second, benchmarks that score agreement with a published review cannot distinguish a model that *read* the primary trial from one that *remembers* the review: the anchor and its primaries are plausibly in training corpora, and contamination inflates apparent competence [Xu et al., 2024, Golchin and Surdeanu, 2024]. Third, the published review is usually treated as an infallible answer key, although hand-made evidence tables carry their own errors: of 27 meta-analyses using standardized mean differences, 17 (63%) contained an extraction error in at least one of the two trials audited against the primary sources, some large enough to negate or reverse a finding [Gøtzsche et al., 2007]; extraction and reporting errors were found in most of the Cochrane reviews examined in an earlier audit, with little effect on their results [Jones et al., 2005]; independent observers extracting the same continuous outcomes produced meta-analyses that disagreed [Tendal et al., 2009], in part because trial reports offer several candidate values for one quantity [Tendal et al., 2011]; and a later methodological review found the evidence base for established extraction standards weak despite that prevalence [Mathes et al., 2017]. A benchmark that cannot notice when the published value is wrong will misgrade its best readers.

EXTRA-I closes those three gaps at once: quantized open-weight models on a consumer machine with no cloud calls; a perturbation reading proof that makes recitation of published values detectable; and a two-layer, source-verified answer key whose quotation-backed rite puts the human key itself on trial. That verification produced a seventeen-entry record on one published anchor, eleven of them discrepancies confirmed against the primary sources, all of it in this study’s public archive [Barra, 2026]. Because the instruments were drafted in the team’s working language before being frozen in English, the study also carries a built-in robustness check on instruction language.

This article reports a pre-registered, five-phase benchmark of local language models as meta-analysis workers — five integrated-GPU models plus a sixth added after the campaign as a pre-registered extension arm — followed by a second campaign, registered before its first call, in which the six run as peers over three anchors and under both extraction sheets, all under contamination-controlled, source-verified grading against the published values.

2 Methods

2.1 Study design and pre-registration

The study protocol — phase order, model cast, hypotheses and compute budget — was registered and committed to the public repository before any run; the two extensions executed within the deployment phase and the post-campaign extension arm that added a sixth model (Section 2.6) were each registered as a dated amendment before its own run. Corpora, sealed perturbations, answer keys, deterministic engines, graders and the detection harness were built and frozen before the campaign, during this study’s own instrument-development phase.

The campaign ran five phases in an ascending line of investigation — P1 *Read*, P2 *Calculate*,

P3 *Create*, P4 *Orchestrate*, P5 *Deploy* — each specified below. A second campaign, registered on 2026-09-08 against a frozen commit and run on 9–10 September 2026, re-ran the reading and creation stages with the six models as peers from the first token, on three anchors and under both extraction sheets; its protocol, instruments and hypotheses are in Section 2.14.

2.2 Anchor meta-analyses and corpora

Two recently published meta-analyses served as anchors. Anchor 1 addresses goal-directed fluid therapy (GDFT) versus conventional fluid therapy in major surgery [Ashraf et al., 2026]; its 14 primary randomized controlled trials constitute the first corpus (8 open-access from PubMed Central; 6 paywalled). It supplies three dichotomous outcome families (overall morbidity, mortality, postoperative ileus) and two continuous ones. The anchor’s own outcome tables roster 4, 2 and 3 of the 14 trials for morbidity, mortality and ileus respectively, and the 14 meet together only in the anchor’s descriptive-characteristics and risk-of-bias tables. All 14 were extracted here, each contributing to whichever outcome families it reports, and the extraction-level analyses span all 14. Anchor 2 addresses low-carbohydrate diets and glycemic control (HbA1c) in type 2 diabetes [Khraise, 2026]; its 7 primary trials constitute the second corpus, and all 7 report its single continuous outcome (change in HbA1c, %), whose published random-effects pooled estimate, -0.24 [$-0.32, -0.16$], is this study’s external reference value. Anchor 3, added by the second campaign as the generalization gate the first campaign’s limitations named, addresses methylene blue in adult shock [Rodríguez-Lima et al., 2026]: a third clinical domain, a dichotomous primary outcome (28–30-day all-cause mortality) published as an odds ratio, OR 0.73 [0.40; 1.36], over eight trials whose per-trial cells exist only inside a raster forest plot; its 8 primaries (5 open-access, 3 paywalled; one in Spanish) constitute the third corpus.

2.3 The perturbation reading proof

Before any model reads a trial, a sealed, deterministic operator displaces load-bearing numbers in its text. A model that returns a displaced value read the text in front of it; one that returns the original published value is reciting training data, a behavior that scales with model size and with a text’s duplication in the corpus [Carlini et al., 2023, Xu et al., 2024]. Models operate exclusively in the perturbed world; the original $\leftrightarrow$ perturbed maps are held by the graders alone and applied only after extraction, as a pure table lookup (the “unperturbation lens”). The sealed maps were reused unchanged across all phases, their SHA-256 fingerprints printed into each phase’s run log. Phase P5 deliberately runs on clean texts and therefore makes no reading-proof claim (Section 2.11).

2.4 Answer keys and the adjudication rite

Each corpus is graded against a two-layer, source-verified answer key: layer 1 holds the anchor’s published cells; layer 2 holds each cell verified against the primary’s full text, with a literal supporting quotation and a per-cell verdict. The key itself is on trial by design: source verification of Anchor 1 produced a public, quotation-backed record of source-confirmed discrepancies in the published meta-analysis (Table 1), and the campaign graded against the key as corrected through the most recent adjudications. Anchor 2’s verification found no discrepancies: its statistics reproduced cleanly.

Table 1: Source-confirmed discrepancies between Anchor 1 and its primary sources, as graded against in this study (confirmed items only; full literal quotations, the withdrawn adjudicator accusations and pending items are in the archived verification record). Issuing formal errata is the journal’s and the authors’ prerogative; conclusion-level impacts are analyzed in the Discussion.

#	Where (anchor)	Published	Source says
1	Characteristics, Yoon	GDFT 36 / control 39	GDFT n=39; control n=36
3	Characteristics, Weinberg ASA	“Not stated”	I–II 7 (27%); $\geq$ III 19 (73%)
5	Characteristics, Sujatha ASA	“95:105” + a spreadsheet-time-corrupted cell	eligibility only (ASA I–II), no distribution
9	Oral-diet outcome, Sun	72 $\pm$ 24 vs 96 $\pm$ 30 h	“shorten ... by 2 days”; medians 4.0 vs 6.0 d
10	Characteristics, de Waal ASA	GDFT 24:123:86:1	inverse: 123 = 52.6% $\times$ 234 (control arm)
11	Characteristics, Diaper ASA	“Not stated”	III–IV 98 (50.0) / 85 (42.9)
12	Flatus outcome, 2 trials	55 $\pm$ 14/58 $\pm$ 16 and 52 $\pm$ 15/60 $\pm$ 18 h	“flatus” absent from both full texts
13	Sample sizes, six trials	presented as randomized	analyzed counts, undeclared
15	Pooled morbidity caption	“Mantel–Haenszel”	numbers reproduce under DerSimonian–Laird
16	Ileus outcome, Castro	ileus 6 (14.0%) / 19 (45.2%)	“ileus” absent; counts are the trial’s PPC result
17	Fluid table, Coeckelenbergh blood loss	GDFT 500 / control 450	the labeled parallel sentences give the inverse

Mechanical grading is backed by an adjudication rite: no cell changes category without the source quotation that decides it, and the adjudicators’ own errors are published in the same record as everyone else’s. The rite exists because judges exhibit a measurable *plausibility bias*: during key construction the graded record reversed this study’s own human–AI adjudicator six times, twice withdrawing accusations of model fabrication in which the models had extracted literally correct values that rigid verification windows had hidden [Barra, 2026]. Quotation-before-verdict is the structural constraint on that bias, for human, hybrid and model judges alike. That adjudication layer is itself the hybrid — the author working with a cloud-based frontier assistant (disclosed in the Declarations) — so the study’s local-only property is scoped to the *evaluated pipelines* (Section 2.6), never to grading and review.

2.5 Instruments

All model-facing instruments are frozen English prompts, committed to the study’s record at registration: the structured extraction sheets, the arithmetic tasks, the calculator protocol, the harness dialogue, and the trigger and synthesis prompts. The instruments were originally drafted in Brazilian Portuguese, the research team’s working language, and translated field-by-field under a published correspondence table (for example, `ic95_md`→`ci95_md`; `RESULTADO`→`RESULT`); the originals are archived, and Section 2.13 describes the robustness analysis this enables.

2.6 Models and hardware

Five quantized open-weight models sized for an integrated GPU formed the comparison cast: `gemma4:12b`, `qwen3:14b`, `llama3.1:8b`, `qwen3.5:9b` and `deepseek-r1:14b`. All phases ran on one consumer mini-PC (AMD Ryzen 7, Radeon 780M integrated GPU, 32 GB RAM, no discrete GPU), served by Ollama with a 16,384-token context, reasoning channels disabled, and a uniform 4,000-token output allowance for extraction tasks; no cloud API was called in any evaluated pipeline (the grading and adjudication layer is cloud-assisted; Section 2.4). One

model was resident at a time, with an explicit memory unload between models. Phase P4 used only `gemma4:12b`, pre-designated in the protocol as the orchestration candidate on the evidence of this study’s instrument-development runs.

After the campaign closed, a sixth model entered as a pre-registered extension arm (protocol Amendment 1, dated 2026-09-07 and committed before its run): `smttek/Qwen3.8-27B:Q2_K_XL`, an openly published mixed-precision quantization of a 27.3-billion-parameter model — the reader the author had meanwhile adopted as primary extractor in a separate systematic-review harness, and the reason for the arm. Its two printed names disagree and neither describes its weights (the tag reads `Q2_K_XL`; the GGUF `file_type` header, which is what the runtime prints, reads `Q4_K_S`), so the build was characterized by reading the tensor table of the weight file itself: 27.32 billion parameters in 10.68 GB at **3.13 bits per weight**, mixed across six ggml types (IQ3_XXS on 77.4% of parameters, IQ3_S on 11.7%, Q3_K and Q2_K on 4.7% each, IQ4_XS on 1.5%, IQ1_M on 0.1%). The cast’s five builds are `Q4_K_M`, about 4.5 bits per weight; resident footprints were 4.9–9.3 GB for the cast and 10.7 GB for the extension arm. Because inference quality below about four bits per weight depends on the quantization scheme and the model size [Dettmers and Zettlemoyer, 2023], the build’s bit depth was measured rather than read from its name. It ran phases P1–P3 under the identical instruments, seals, keys, engines and graders, called exactly like the cast (same context, reasoning off, same output allowance), 54 calls in 119 minutes. After the graders were re-run over the extended cast, the five-model record was verified unchanged field by field.

2.7 Phase P1 — structured extraction

Each of the five models, and later the extension arm, extracted all 14 perturbed Anchor-1 primaries from zero, twice (two independent replicates), under the frozen English extraction sheet: 30 fields per trial, each an object `{"value", "where"}`, the mandatory **where** being the model-side provenance instrument — a locator naming the section or table the value came from, not a verbatim quotation. Grading used the eligible cell set of the two-layer key — cells with a usable, digit-bearing source-verified value, 124 per model — comparing the seal-reversed first-parseable sheet against the key’s source layer with a magnitude-based numeric comparator. That comparator is a declared approximation of the full ruler, so every non-matching cell was listed for adjudication rather than auto-verdicted. Replicate stability applied the same comparator between the two raw replicates. Recitation candidates were detected mechanically — a cell whose raw text contains a seal pair’s original value while the displaced image the model actually read is absent — and each was adjudicated against the perturbed source text before any verdict. Divergent cells were classified by factuality (Table 2 defines the three classes) as *omission*, *invention candidate* — a number absent from the perturbed text and not derivable from it by percentage, sum, difference or mean — or *value/format divergence*; the same invention screen ran over every cell of the Anchor-2 sheets of phase P3, both replicates. Each non-matching cell was then re-examined against the perturbed text its model read and classified by mechanism, with residues adjudicated under the rite; the cell-by-cell record is Supplementary Tables S1–S7.

2.8 Phase P2 — meta-analytic arithmetic

Each model computed, over its own P1 sheets, per-study risk ratios and mean differences with 95% confidence intervals, and pooled estimates, in two arms: arm A unaided, and arm B through a plain-text calculator protocol in which the model writes lines of the form `CALC: function(arguments)`, the harness executes the validated function and returns `RESULT:` into the context (up to 5 rounds and 20 calls) — a text-only form of tool use [Schick et al., 2023, Gao et al., 2023] — and the model then answers with a final JSON. Three prompt families (risk ratios, mean differences, pools) $\times$ two arms $\times$ two replicates ran per model. Grading

recomputed the truth with the same validated engine from the input block the model saw, under pre-registered parsing rules: event counts read from forms such as “19 (32.8%)”; percent-only cells converted against the group size; means accepted only as mean $\pm$ SD, so median (IQR) cells have no computable truth and *not computable* is correct there. Each quantity was labeled exact, right-direction, wrong, correctly-refused, wrongly-refused, or unsupported; intervals and pooled estimates were graded separately.

2.9 Phase P3 — deterministic creation of both meta-analyses

Everything downstream of the sheets was executed by pre-registered, frozen code; no model touched a number after extraction. For Anchor 1, the engines parsed each model’s P1 sheets, selected event/denominator routes per outcome, computed per-study risk ratios, pooled them under both Mantel–Haenszel and DerSimonian–Laird [DerSimonian and Laird, 1986], applied the sealed reversal, and classified every difference from the published tables into a frozen five-category comparison (*reproduces / differs by a documented anchor discrepancy / model route / model error / source unavailable*). The pooled comparison uses DerSimonian–Laird because the published pooled numbers instantiate that method under a Mantel–Haenszel caption (documented discrepancy #15). For Anchor 2, each model freshly extracted the 7 perturbed primaries (two replicates) under the English continuous-outcome sheet, which, unlike the Anchor-1 sheet, records bare values with no provenance field — an instrument asymmetry carried over from this study’s instrument-development draft of the same sheet. A deterministic route selector then built each trial’s sextet (mean, SD, n per arm) by the frozen rules: declared change used directly; standard errors and confidence intervals converted to SDs; baseline-and-final pairs combined under the Cochrane $r = 0.5$ rule [Higgins et al., 2023]. The DerSimonian–Laird pool was then computed twice: in the perturbed world, and through the graders’ sealed lens for comparison with the published $-0.24 [-0.32, -0.16]$. Tolerances were pre-registered at ± 0.01 for risk ratios and interval bounds and ± 0.1 for mean differences.

2.10 Phase P4 — orchestration under a warn-only detection harness

The study’s ten-net detection harness follows the doctrine that nets *detect and warn, never substitute a value*. The doctrine is empirical: in this study’s instrument-development testing, checking stages given write-power did net harm through the same plausibility bias — an auditing model “repaired” correct cells by imputing plausible values (once correcting a sample size to 24.5 while quoting, in its own verdict, the text’s “n = 28”), a calculator stage flipped the sign of a control arm that had genuinely worsened (+0.3 written as -0.3), and a pipeline’s quality gate did six times more damage to the pooled estimate than the planted sabotage it caught [Barra, 2026]. The harness comprises schema-constrained typed calls (`{"function", "arguments", "source", "derivation"}`), an arity check with the full function signature, a sign-echo net, an arm-agnostic declared-source check, a declared-derivation check, an argument type system with per-class value impossibilities, a confidence-interval bound-order check, a coherence check tying each interval call to the study’s own last executed mean difference, a negative-SD result warning, a closing-versus-executed check, per-call warning budgets with confirm-by-repeat, and an escalation flag (`requires_human_review`) on insistence. `gemma4:12b` then ran one formal pipeline over its own Anchor-2 sheets: per-study typed calls, a model-emitted DerSimonian–Laird pooling call over its own sextets, a product-layer coherence check on the reported intervals (flagged, never endorsed), and a narrative synthesis checked mechanically for orphan numbers. No model stage saw an answer key, a published value, or the seal; grader-side comparison happened only afterward.

2.11 Phase P5 — deployment on clean texts

The deployment phase ran the same architecture on the *original, unperturbed* texts of both anchors with no answer key in any loop — the day-to-day usage scenario. Its components: (i) fresh two-replicate extraction of all 21 primaries by `gemma4:12b` under the English instruments; (ii) a frozen discrepancy-cell panel — the seven of Anchor 1’s eleven source-confirmed discrepancies that resolve to a single extractable cell with a decided source value — testing whether the model’s cell sides with the source or with the anchor’s published value; (iii) deterministic recomputation of both meta-analyses and per-trial three-way tables (source quotation $\times$ the anchor authors’ cell $\times$ the model’s cell); and (iv) a three-configuration comparison of error detection over the same sheets: the deterministic engine alone, the engine plus deterministic sheet-versus-source nets, and the model orchestrating under the conversational ten-net harness (Table 6).

2.12 Comparison conventions and software

All computation used Python 3.12 with the repository’s validated engine functions (risk ratio and its log-scale interval; mean difference and its interval; SD conversions; Mantel–Haenszel, DerSimonian–Laird and inverse-variance pooling); the risk-ratio function reproduces a published per-study value of Anchor 1 as its standing test case. No external statistics package was used. Replicates measure stability, not statistical significance; with one corpus per anchor the model comparisons are descriptive. Compute totaled 277 model calls across P1–P4 (≈ 10.8 h of machine time) plus 54 calls (≈ 2.0 h) for the extension arm, every queue resume-safe.

2.13 Robustness analysis: instruction language

This study’s instrument-development phase had exercised the identical corpora, seals, keys, engines and graders under the Portuguese drafts of the instruments (Section 2.5), so that archived record provides a natural comparison arm in which instruction language is the only changed variable. For each phase, the directly comparable quantities of the two records were compared, under the pre-registered acceptance criterion that differences stay within replicate-level variance.

2.14 The six-model, three-anchor campaign

The first campaign added its sixth model after the fact and measured the mandatory-quotation sheet on four models and two anchors. The second campaign removed both asymmetries by design, under a protocol registered on 2026-09-08 against a frozen commit and reviewed before registration by five independent lenses, each finding put to a skeptic (77 raised, 34 refuted, 43 applied). *Cast*: the same six builds (Section 2.6), peers from the first token, one resident at a time in cast order. *Anchors*: the two above and Anchor 3. *Sheets*: both frozen English extraction sheets ran as a declared arm on every anchor — v1, the locator-only sheet of the first campaign, and v2, which makes a verbatim source quotation mandatory for every cell (`{"value", "where", "quote"}`), the instrument the first campaign’s own A/B had measured on four models and two anchors and declared unsettled at that scale. *Calls*: $6 \times (14 + 7 + 8) \times 2 \times 2 = 696$, two replicates per cell, perturbed corpora only. *Execution*: because Anchor 3’s largest primary needs 14,741 prompt tokens, the context was raised from 16,384 to 24,576 tokens for every model and anchor, and the change was registered as a variable with its own control (H12.5); reasoning off, vendor sampling, output allowance 4,000 tokens (v1) and 8,000 (v2), no seed — runs are not bit-reproducible, and replicate stability is the instrument for that. The harness was built resumable by construction: an output is written atomically and accepted only if it parses, carries the expected keys and did not stop on the token ceiling; a file failing any check is set aside with its reason; every call records tag, anchor,

sheet, trial, replicate, token counts, stop reason, duration and the SHA-256 of its prompt; and an inventory counts what exists against the 696 the protocol enumerates.

Anchor 3’s key and instruments. The review prints no per-trial cells in text; they were recovered from its forest plot by vision and verified by reconstruction — the sixteen read cells predict the 48 other values printed on the figure (eight odds ratios, sixteen limits, eight weights, τ^2 , χ^2 , I^2 and the diamond), and all 48 reproduce — so layer 1’s provenance is a figure coordinate plus that arithmetic, the first key of the series with no transcribable sentence behind its published cells. Layer 2 verified all eight trials against their sources with quotations: six agree cell for cell; two do not (Table 9), and the corrected cells were committed before any run, together with the review’s internal contradictions and the frozen handling of each. Grading is against layer 2, so no verdict rests on an unquoted cell, and no model is ever graded against the published 0.73. Two pooling targets were declared because they answer different questions: the campaign engine pools each model’s cells under DerSimonian–Laird, which measures the model; the published diamond does not reproduce under that estimator (0.737 [0.449; 1.208]) but does, digit for digit, under Paule–Mandel with the Hartung–Knapp adjustment and Student’s t on $k - 1$ degrees of freedom [Veroniki et al., 2016, IntHout et al., 2014] — a method the review’s methods section does not name — so that route is run by the comparator alone, to measure the review. The anchor-3 sheets ask, for every arm the trial has, the event count, the denominator, the reported percentage and the outcome timepoint; they carry no polarity instruction and do not reduce multi-arm trials — the reduction (both active arms against the shared control, 15/60 against 14/30 for the three-arm trial, the combined-arms rule of the Handbook [Higgins et al., 2023]) is the corrector’s, under a rule frozen in the protocol — and one trial’s 21-day figure enters an analysis the review labels 28–30 days, with its exclusion run as a pre-registered sensitivity. The departure from the anchor-1 sheet is declared: four fields per arm over enumerated arms are the two axes on which the review’s human extractors failed, so a model that lands the errata is helped by the instrument, and the finding is read against that sentence.

The reading proof on Anchor 3. Its cells are counts, and no integer event count is perturbable: a one- or two-digit integer displaced by 5–15% lands on another small integer that occurs dozens of times in any clinical article. The proof was therefore placed on the per-arm denominator, displaced with every dependent percentage recomputed for coherence — a new capability of the sealed operator, proved before freezing (two earlier versions failed the proof and were discarded) — and graded through a boundary-aware single-pass lens built for this anchor alone, because on these texts the frozen lens commits 42 undue substitutions and the new one none. Under the coherence constraints the design covers 10 of 17 arms and four and a half of eight trials; the event counts carry no reading proof, and three trials none at all. Every anchor-3 accuracy figure is read against that denominator. Independently of the seal, one recitation trap exists at no cost: a model that emits the review’s 24/30 for the inverted trial has reproduced the review rather than read the primary.

Hypotheses, registered. H12.1, the generalization gate: every model’s anchor-3 pool is digit-consistent with its own cells within ± 0.01 . H12.2, the decisive hypothesis: of six models, how many produce the corrected Aguilar cell (6/30, not the review’s 24/30) and enumerate Shaker’s three arms so that the frozen reduction yields 15/60 — ≥ 1 of 6 on both cells was pre-registered as the finding that a local reader recovered what peer review missed, and 0 of 6 as the null. H12.3, the sheet A/B at scale: v2 preserves accuracy (mean v1→v2 change on anchor 1’s 124 cells no worse than -4) and buys estimate quality (closer to layer 2 in ≥ 4 of 6 models on ≥ 2 of 3 anchors). H12.4, exploratory: the provenance nets’ flag rates on the table-dwelling anchor. H12.5, the context control: gemma4:12b’s anchor-1 v1 agreement within

103 ± 4 cells of the first campaign and its anchor-2 lens within ± 0.05 of -0.27 ; outside those bands the context change is declared a confounder before any anchor-3 result is read. H12.6, the null, bounded in advance: no model above 50% of the 32 anchor-3 cells.

Grading and adjudication. Phase 2 graded every sheet mechanically — the frozen comparator on anchors 1 and 2; exact integer comparison on anchor 3, because the frozen comparator’s zero rule would accept 10, 20 or 30 as 0 — and Phase 3 ran the frozen engines. Phase 4 took every non-matching cell to adjudication under the rite, in the mechanism classes of Supplementary Tables S1–S7 extended to anchors 2 and 3, with a reader’s quotation-bound verdict for whatever no rule decided. The adjudicated score adds only the cells in which the model was right and the grader wrong (the 2026-09-01 rule); faithful re-encodings, population-layer choices and summaries stay divergent, and cells that put the key on trial are listed for the author, not flipped. The adjudication was performed by the assistant under the author’s delegation, every verdict a row the author can overturn.

3 Results

3.1 P1 — extraction: the reading proof held, and no attributable recitation occurred

All 140 extraction calls ($5 \text{ models} \times 14 \text{ trials} \times 2 \text{ replicates}$) completed with clean stops in 390.4 minutes; every sheet parsed at the sheet level except one replicate (**deepseek-r1:14b**, one trial), whose pair served. Cell agreement with the corrected two-layer key — after the four grader-artifact adjudications (Table 2 caption) — ranged from 89/124 (71.8%) to 105/124 (84.7%), and replicate stability from 77.4% to 99.2% (Table 2). The extension arm’s 28 sheets completed with clean stops in 77.9 minutes and all parsed; graded identically, it scored 108/124 (87.1%) — the highest of the six (Table 2, last row). Because the numeric comparator cannot see format or granularity equivalence, cross-model ranking by this score alone is provisional; the value-level separation appears in P3. The same ruler applied to the anchor’s own published cells — the human extraction, layer 1 against layer 2 — returns 102/124 (82.3%; Table 2, first row): twelve of its 22 non-matching cells are the source-confirmed discrepancies of Table 1, and the other ten fall into the classes the models’ divergences fall into.

The mechanical recitation detector raised 5 candidate cells across the cast’s 140 sheets, and adjudication against the perturbed source texts resolved all 5 to incomplete-perturbation artifacts rather than recall: three models independently wrote a displaced count (113) that is arithmetically reconstructible from a surviving percentage printed beside it in the perturbed text ($57.7\% \times n=196 = 113.1$), and two models transcribed a value (4088) that survives verbatim in a prose restatement the perturbation operator missed. Zero attributable recitations remained. The same three models copied, rather than reconstructed, the equally inconsistent control-arm cell of the same table — an asymmetric preference shared across architectures, recorded as a behavior note.

The factuality classification of the divergent cells (Table 2, middle columns) found **zero invention candidates in every model**, the extension arm included. Most divergences were themselves replicate-stable: 17/21 for **gemma4:12b**, 20/20 for **qwen3:14b**, 13/20 for **qwen3.5:9b**, 16/26 for **deepseek-r1:14b**, but only 17/36 for **llama3.1:8b**, and 16/17 for the extension arm (counted over the comparator’s raw divergent inventories, before the four-cell adjudication; **deepseek-r1:14b**’s denominator excludes the divergent cell in its unparseable-replicate trial).

Supplementary Tables S1–S7 (after the references) account cell by cell for the 124 comparator failures across the five models — the divergences behind these counts, not a per-model total — each re-examined against the perturbed text its model read and classified by mechanism: 12 omissions; 27 equivalent re-encodings; 26 population-layer choices; 34 case-mix summaries; 12

genuine reading slips; 9 unmatched ratio-style re-expressions; and 4 cells adjudicated *in the models’ favor* as a grader-side lookup collision (Table S7). The extension arm’s 17 divergent cells fall into the same mechanisms: the seven case-mix summaries every reader produces, the two FEDORA population-layer cells, one re-encoding (0 (0) for 0 (0%)), six omissions and the same Castro collision cell.

Table 2: P1 extraction under English instruments (perturbed Anchor-1 corpus; 124 eligible key cells per model). Two comparators: *stability* grades replicate 1 against replicate 2 (the model against itself, no key); *correct* grades replicate 1 against the source-verified key. The three middle columns decompose the non-matching cells by factuality (omissions + inventions + value/format = 124 – correct): *omission* = the model wrote *not reported* where the key holds a value; *invention* = a number absent from the source text and not derivable from it; *value/format* = content that traces to the source but differs from the key by value route (e.g. a different population layer) or by format of the same information (e.g. percentages for counts, a category summary for an enumeration). A cell can be stable yet divergent from the key, so stability can exceed the key score. Recitation candidates before/after adjudication. †one replicate unparseable (its pair served). One value/format cell each for **qwen3:14b**, **qwen3.5:9b**, **deepseek-r1:14b** and **llama3.1:8b** was adjudicated as a grader-side artifact — the transcription was verbatim-correct and a seal-pair substring collision in the unperturbation lookup corrupted the comparison (Supplementary Table S7); the four affected rows report the adjudicated scores (the comparator’s raw output was 104, 104, 97 and 88 correct, with value/format 14, 16, 25 and 36, preserved in the archived record). The last row, below the rule, is the post-campaign extension arm (Section 2.6), graded identically; its Castro collision cell was adjudicated by the same rule (raw 107 correct, value/format 11). The first row, above the rule, is the anchor’s own extraction — layer 1, the cells the review published — scored on the same ruler against layer 2: its 22 non-matching cells are 4 omissions (“not stated” where the source prints a distribution), 2 inventions (Table 1, #9: hours printed nowhere in the source) and 16 value/format cells (swapped arms, a conversion of medians, rounding, a summary, a definitional count); twelve of the 22 are the source-confirmed discrepancies of Table 1. Stability and recitation do not apply to it.

Extraction	Stability	Recit.	Omissions	Inventions	Value/format	Correct (vs key)
Published (human)	—	—	4	2	16	102/124 (82.3%)
gemma4:12b	119/124 (96.0%)	1 → 0	0	0	21	103/124 (83.1%)
qwen3:14b	123/124 (99.2%)	1 → 0	6	0	13	105/124 (84.7%)
qwen3.5:9b	103/124 (83.1%)	0	4	0	15	105/124 (84.7%)
deepseek-r1:14b†	96/118 (81.4%)	1 → 0	2	0	24	98/124 (79.0%)
llama3.1:8b	96/124 (77.4%)	2 → 0	0	0	35	89/124 (71.8%)
qwen3.8:27b (ext.)	117/124 (94.4%)	0	6	0	10	108/124 (87.1%)

3.2 P2 — arithmetic: unaided fails, the calculator repairs

Across the five-model cast, exactly 1 of 57 unaided 95% confidence intervals was exact (arm A); through the text-protocol calculator (arm B), exact intervals rose to 41/50 (82%) among the four models that operated the tool, and exact point estimates rose in step (Table 3). The extension arm reproduced the unaided signature (0/16 exact intervals) and, with the calculator, closed every quantity its sheets support — points, intervals and, uniquely among the six, the pooled estimates — with no malformed output (Table 3). This phase has no human row in Table 3, because the reviews supply none: their arithmetic is software arithmetic, and the frozen engine reproduces each published pooled estimate from the published cells digit for digit under the estimator the review actually used — Anchor 1 under DerSimonian–Laird despite its Mantel–Haenszel caption (Table 1, #15), Anchor 2 under DerSimonian–Laird, Anchor 3 under Paule–Mandel with Hartung–Knapp, unnamed in its methods (Table 9, #5). The human reference for arithmetic is therefore exact by construction, and the table compares the models with the engine.

Table 3: P2 arithmetic, per model: exact quantities unaided (arm A) versus through the text-protocol calculator (arm B). No human row: the reviews’ own arithmetic reproduces exactly from their cells under the frozen engine (see text). Across the cast, 1 of 57 unaided intervals was exact, and 41/50 (82%) of arm-B intervals among the four models that operated the tool; pooled estimates stayed at or near zero exact in both arms for the whole cast, and only the extension arm (last row, below the rule) closed them, 12/12. †`deepseek-r1:14b` is not among the four models that operated the tool: its unaided arm was unusable (5 of 6 outputs unparseable, zero gradable quantities), and its pooled-family tool arm ran without emitting a single **CALC**: line.

Model	Exact point estimates (A $\rightarrow$ B)	Exact intervals (arm B)
<code>gemma4:12b</code>	4 $\rightarrow$ 12	11/16
<code>qwen3:14b</code>	4 $\rightarrow$ 14	14/16
<code>llama3.1:8b</code>	5 $\rightarrow$ 8	8/9
<code>qwen3.5:9b</code>	5 $\rightarrow$ 8	8/9
<code>deepseek-r1:14b</code> †	—	—
<code>qwen3.8:27b</code> (ext.)	6 $\rightarrow$ 16	16/16

3.3 P3 — creation: every model reproduces the dichotomous pools; one of the cast, and the extension arm, reach the continuous diamond

From the P1 sheets, the deterministic engines reproduced Anchor 1’s published pooled estimates for *all five* models and for the extension arm: pooled morbidity 0.771–0.787 and mortality 0.980–1.023, every value within ± 0.01 of the published except one mortality estimate at $\Delta 0.04$ (Table 4). The extension arm’s morbidity pool sits farthest from the cast’s: on FEDORA it read the literal event counts (18/224 and 35/226) and reproduced the published per-study risk ratio 0.519 exactly, where `gemma4:12b` read the percentages and derived 19/38 — a route difference worth $\Delta 0.008$ at the pool. The postoperative-ileus pool is non-comparable by construction (documented discrepancy #16) and was reported, never pooled against: over the two genuinely-ileus trials, the pool was 0.821 [0.559, 1.207] under Mantel–Haenszel and 0.429 [0.048, 3.87] under DerSimonian–Laird — both crossing 1 — identically from the perturbed and the clean-text extractions.

Table 4: P3, Anchor 1: DerSimonian–Laird pools from each model’s own sheets, against the published diamonds (the human extraction, as printed) and layer 2 (the same engine over the source-verified cells). Every model contributed every trial of the anchor’s own per-outcome roster (4/4 for morbidity, 2/2 for mortality — a meta-analysis pools, per outcome, only the trials reporting it; Section 2.2), so no trials-in-pool column is needed here — unlike Anchor 2 (Table 5), where sheet starvation varies by model.

Extraction	Morbidity RR [95% CI]	Mortality RR [95% CI]
Published (human)	0.778 [0.567, 1.068]	1.021 [0.446, 2.337]
Layer 2 (source)	0.787 [0.580, 1.068]	1.023 [0.446, 2.346]
<code>gemma4:12b</code>	0.779 [0.569, 1.065]	1.023 [0.447, 2.344]
<code>qwen3:14b</code>	0.779 [0.569, 1.065]	1.021 [0.446, 2.337]
<code>llama3.1:8b</code>	0.771 [0.562, 1.059]	1.023 [0.447, 2.344]
<code>qwen3.5:9b</code>	0.779 [0.569, 1.065]	1.019 [0.445, 2.336]
<code>deepseek-r1:14b</code>	0.775 [0.565, 1.063]	0.980 [0.428, 2.246]
<code>qwen3.8:27b</code> (ext.)	0.787 [0.580, 1.068]	1.023 [0.446, 2.346]

From the fresh Anchor-2 extractions (70/70 calls, 155.1 minutes; the extension arm’s 14/14 in 27.6 minutes), the same engine and sealed lens produced the cast’s five diamonds and the extension arm’s sixth (Table 5). `gemma4:12b`’s lens landed at -0.27 and the extension arm’s at -0.25 [-0.33 , -0.16], against the published -0.24 [-0.32 , -0.16] — the closest of the six;

both fall within the pre-registered ± 0.1 tolerance at the point estimate and at both bounds, and the extension arm’s reconstructs the anchor’s own homogeneity (Table 5). The other four landed between -0.47 and -0.80 with saturated heterogeneity, three of them from starved sheets. The invention screen over all 1,104 Anchor-2 sheet cells (five models, both replicates) found zero invention candidates in any model; a re-implementation of the same rule over the extension arm’s 224 Anchor-2 cells (196 numbers) found none either, and returned zero for the five as well.

Table 5: P3, Anchor 2: heterogeneity anatomy of the cast’s five diamonds, and the extension arm’s, against the published (human) extraction — pooled HbA1c mean differences through the unperturbation lens, i.e. each model’s sheet with the graders’ sealed number-map reversed by pure table lookup before pooling (Section 2.3); models extracted in the displaced world and never saw the originals, and the lens corrects nothing — reading errors pass through it intact. The published row is as the anchor prints it (I^2 6%, τ^2 0.00). The last column is the same pool *before* reversal, in the displaced world; it demonstrates the correct-nothing property mechanically — reversal collapses `gemma4:12b`’s I^2 from 91.3% to 28.6% and the extension arm’s from 91.2% to 7.1% (in faithful sheets the perturbation displacement *was* most of the heterogeneity) while leaving the other four models’ I^2 unchanged to within 0.1 points (heterogeneity injected by reading survives the lookup by construction).

Extraction	Trials in pool	Lens MD [95% CI]	τ^2	I^2	I^2 before reversal
Published (human)	7/7	$-0.24 [-0.32, -0.16]$	0.00	6%	—
qwen3.8:27b (ext.)	7/7	$-0.25 [-0.33, -0.16]$	0.001	7.1%	91.2%
gemma4:12b	7/7	$-0.27 [-0.38, -0.17]$	0.005	28.6%	91.3%
deepseek-r1:14b	7/7	$-0.47 [-1.22, 0.28]$	0.997	99.1%	99.1%
llama3.1:8b	5/7	$-0.60 [-1.02, -0.18]$	0.199	90.5%	90.4%
qwen3:14b	6/7	$-0.63 [-1.08, -0.17]$	0.291	95.2%	95.2%
qwen3.5:9b	6/7	$-0.80 [-1.67, 0.07]$	0.898	95.0%	95.1%

3.4 P4 — orchestration: confirm-or-correct resolved both warnings, and the pooling call closed digit-consistent with the executed result

`gemma4:12b` closed all 7 per-study call sequences under the English ten-net harness in 8.7 minutes. Two content warnings fired — both on one trial, both from the declared-derivation net — and confirm-or-correct resolved both; no other net fired and no format failure occurred. The model then emitted the pooling call over its own seven sextets and closed digit-consistently with the executed result ($-0.50 [-0.78, -0.22]$); the mechanical truth over the same sheets was $-0.52 [-0.82, -0.22]$ (Δ MD 0.02, from two named route divergences). The product layer flagged zero incoherent intervals, and the synthesis contained zero orphan numbers. The grader-side lens over the same sheets returns `gemma4:12b`’s P3 diamond (Table 5).

3.5 P5 — deployment on clean texts: the discrepancies are visible, and the one miss is one cell

On the original, unperturbed texts, replicate agreement was 96.8% (Anchor 1) and 95.4% (Anchor 2), and all 42 sheets parsed. At every cell of the frozen discrepancy panel — 7 of 7, including all four direction-critical swaps — the model’s extraction sided with the *source* against the anchor’s published value; on the trial of discrepancy #16, the model wrote *not reported* in both replicates rather than reciting the published numbers. The deterministic recomputation reproduced Anchor 1’s pooled morbidity (0.779 [0.569, 1.065]) and mortality (1.019 [0.445, 2.336]) within tolerance; the Anchor-2 pool missed its tolerance at $-0.34 [-0.51, -0.18]$, and the residue decomposed entirely to one cell — the model transcribed a printed “ -1.6 ± 0.3 ” from one trial’s results prose and declared the $\pm$ a standard deviation, where the value equals the half-width of the confidence interval printed in the same trial’s table.

Correcting that single cell by the key’s own conversion rule yields -0.24 $[-0.32, -0.16]$ (I^2 3.7%), the published diamond to the hundredth; the cell was identical in both replicates, and zero invented values occurred in the phase’s 124 graded cells.

Over the same sheets, the three detection configurations behaved differently against that poisoned cell (Table 6). The bare engine delivered the distorted pool silently. The sheet-versus-source net flagged the experimental arm only after a first rule revision: its initial form stayed correctly silent because the trial’s own prose prints the $\pm$ form, so the rule was extended, as a dated amendment, to fire when a printed $\pm$ equals the printed interval’s half-width (the control arm’s $\pm$ matches nothing within the pre-registered tolerance and was not flagged). The conversational ten-net harness raised zero flags on the poisoned cell — whose downstream calls were well-formed, correctly source-declared and internally coherent — while catching four live orchestration slips (one arity, three derivation declarations) in the same run. A DerSimonian–Laird weight-share flag ($>40\%$) was blind too: the poisoned trial’s share was 19.3%, and the distortion surfaced as heterogeneity inflation (I^2 6% $\rightarrow$ 79%).

Table 6: P5: three error-detection configurations against the same single-cell defect, same sheets, no answer key in any loop.

Configuration	Pool delivered	The poisoned cell
(a) deterministic engine, no nets	-0.34 $[-0.51, -0.18]$	silent
(b) engine + sheet-vs-source nets	-0.34 (unchanged; warn-only)	flagged (exp. arm, with quotes)
(c) model + conversational ten-net harness	-0.32 $[-0.46, -0.19]$	not flagged; 4 live slips caught

3.6 Instruction-language robustness: every phase within replicate-level variance

Per phase, the campaign’s results sat within replicate-level variance of this study’s own instrument-development record, in which the identical materials had run under the Portuguese drafts (Table 7). The largest reading-stage difference was +3 cells of 124 on the one directly comparable model, with replicate stability identical to the decimal; the arithmetic signature (unaided failure, calculator repair, pool ceiling, per-model deviant classes) matched point for point; the creation-stage flagship lens was identical in point estimate *and* interval; and the orchestration-stage delta sat inside the measured route-variability band with the same warning classes on the same trial.

Table 7: Instruction-language robustness: the campaign (English instruments) vs this study’s own instrument-development record (Portuguese drafts), per phase (directly comparable quantities only).

Phase	Campaign (English)	Development record (Portuguese)
Read (gemma4:12b, cells / stability)	83.1% / 96.0%	80.6% / 96.0%
Calculate (unaided exact CIs; tool arm)	1/57; 82% exact	0/30; same signature
Create (gemma4:12b lens, Anchor 2)	-0.27 $[-0.38, -0.17]$	-0.27 $[-0.38, -0.17]$
Orchestrate (pool vs truth; behavior)	$\Delta 0.02$; confirm-or-correct	$\Delta 0.06$; same classes

3.7 Six models, three anchors, both sheets: the registered replication and extension

All 696 calls completed and passed the integrity gate — zero refusals, zero truncations — in 39.7 h of summed call time (3.42 min per call; 7.52 million input and 0.78 million output tokens;

the largest prompt 17,898 of 24,576 tokens). The wall clock was about 28 h, during roughly eight of which a second harness instance, launched by hand into the same queue, competed for the integrated GPU: calls took twice as long and nothing was corrupted, the atomic writes and the resume gate having been built for exactly that; a single-instance lock was added afterwards, before any grading, as a dated instrument fix. Table 8 gives every model’s cell agreement on the three anchors under both sheets, and Tables 10 and 11 the pools the frozen engines built from those sheets.

Table 8: The six-model campaign: cells agreeing with the source-verified key, per anchor and sheet (adjudicated scores; the comparator’s raw output, one cell lower wherever a Castro collision or an anchor-2 lens miss was adjudicated in the model’s favor, is preserved in the archived record). Anchor 1: 124 eligible cells; Anchor 2: 49; Anchor 3: 32 — with the reading proof covering four and a half of its eight trials. †sheets refused by the strict JSON reader because a verbatim quotation carried a control character inherited from the PDF-derived source text; a tolerant reader, run as a declared sensitivity, recovers them to 28/32, 20/32 and 30/32 respectively. The first row is each anchor’s own extraction — layer 1, the cells the review published — scored on the same ruler against layer 2 (Anchor 1: 12 of its 22 non-matching cells are the discrepancies of Table 1; Anchor 2: the verification found none; Anchor 3: the three cells of Table 9, items 1–2).

Extraction	Anchor 1 (GDFT)		Anchor 2 (low-carb)		Anchor 3 (methylene blue)	
	v1	v2	v1	v2	v1	v2
Published (human)	102/124		49/49		29/32	
gemma4:12b	109/124	106/124	33/49	31/49	23/32	23/32
qwen3:14b	105/124	85/124	24/49	21/49	32/32	24/28†
llama3.1:8b	92/124	86/124	24/49	16/42	26/32	23/32
qwen3.5:9b	95/124	91/124	30/49	25/49	28/32	16/28†
deepseek-r1:14b	109/124	102/124	25/49	24/49	32/32	22/24†
qwen3.8:27b	111/124	110/124	35/49	31/49	28/32	28/32

Anchor 3: cells and pools (H12.1, H12.2, H12.6). No sheet of any model reproduced either of the review’s two errata — zero cells agreeing with the published 24/30 or 15/30 against the source (Table 11). On the v1 sheet, four of six models read Aguilar as 6/30 from the Spanish primary (**qwen3:14b**, **llama3.1:8b**, **qwen3.5:9b**, **deepseek-r1:14b**); the other two returned no mortality entry for that trial at all, on both sheets, and were graded as omissions rather than as errors. Six of six enumerated Shaker’s three arms so that the frozen reduction yielded 15/60, and four of six landed both cells. Two models, **qwen3:14b** and **deepseek-r1:14b**, scored 32 of 32 cells and pooled, under the campaign’s engine, exactly to the layer-2 constant, OR 0.484 [0.319; 0.735], and under the review’s own estimator to 0.484 [0.342; 0.685] — the corrected diamond, significant, where the review published 0.73 [0.40; 1.36] and concluded that the randomized evidence shows no mortality benefit; on the same ruler the review’s own cells score 29 of 32, and only those two perfect sheets score above them (Table 11). The pre-registered null (H12.6) fell in eleven of the twelve model–sheet pairs. Every model’s pool was digit-consistent with its own cells (H12.1), but the parse gate failed on three v2 sheets whose mandatory quotation carried the source’s control characters — an instrument finding about PDF-derived text, not about reading, recorded as a declared sensitivity (Table 8). Of the 63 non-matching cells on this anchor, 56 are omissions (five sheets wrote nothing for the Spanish trial; four for the trial that publishes no per-arm count, which the key tallied patient by patient), and the seven reading slips are two survivor counts reported as deaths, two percentages written in the denominator slot, and three cause-specific partial counts from a trial that publishes survivors.

Table 9: Source-confirmed discrepancies and internal inconsistencies of Anchor 3 [Rodríguez-Lima et al., 2026], registered before any model call, each with its deciding quotation in the archived record. Items 1 and 2 were confirmed against the primary sources; 3–6 are verifiable inside the published article. By the review’s own estimator the published diamond, OR 0.73 [0.40; 1.36], becomes 0.60 [0.33; 1.12] with item 1 corrected, 0.59 [0.39; 0.91] with item 2, **0.48 [0.34; 0.69]** with both, and 0.49 [0.33; 0.74] with both corrected and Aguilar removed altogether: the published conclusion survives neither correction by any route. Unlike Anchor 1, where a discrepancy was first raised by the local models, every item here was raised by the adjudicator during key construction, before any model read the anchor; the models were measured against these errata, not credited with finding them. Issuing errata is the journal’s and the authors’ prerogative; an alert table has been drafted and held.

#	Where (anchor)	Published	Source says
1	Forest plot, Shaker 2025	15/30 in the methylene-blue arm	three arms (placebo 14/30; MB 1 mg/kg 9/30; MB 4 mg/kg 6/30): the numerator sums both active arms, the denominator keeps one; correct reduction 15/60 vs 14/30 — trial OR 1.14 [0.41; 3.15] → 0.38 [0.15; 0.96]
2	Forest plot, Aguilar 2016	24/30 and 19/30	mortality 20.0% and 36.6% in arms of 30, i.e. 6 and 11 deaths; the published cells are the survivors — trial OR 2.32 [0.72; 7.41] → 0.43 [0.14; 1.38]; the review’s own Table 1 records “↓ Mortality” for this trial
3	§3.5 text vs forest plot	“restricted to the eight septic shock trials, excluding Levin”	the plotted eight include Levin 2004 and exclude Kuri 2025
4	GRADE table vs forest plot	8 studies, $n = 479$	the plotted arms total $213 + 218 = 431$; 479 is a third composition
5	Methods vs printed diamond	random effects, estimator unnamed	does not reproduce under DerSimonian–Laird (0.737 [0.449; 1.208]); reproduces digit for digit under Paule–Mandel with Hartung–Knapp
6	Outcome label vs Aguilar	28–30-day mortality	the trial reports mortality at ICU discharge and 21 days and states that at 28 days no difference was found, giving no numbers

Table 10: The six-model campaign, Anchors 1 and 2: pools built by the frozen engines from each model’s own sheets, per sheet (estimate and 95% confidence interval), against the published values (the human extraction, as printed) and layer 2 (the source-verified cells under the same engine). Anchor 1: DerSimonian–Laird risk ratios (layer 2: morbidity 0.787 [0.580; 1.068], mortality 1.023 [0.446; 2.346]; published 0.778, 1.021). Anchor 2: the mean-difference lens, with the number of trials it could pool in parentheses (layer 2 and published: -0.24 [-0.32 ; -0.16]). ‡one stray denominator — 31, a figure-caption number written as de Waal’s control-arm n — took the mortality pool to 0.40 [0.02; 9.12]; the same sheets reproduced the morbidity pool exactly.

Extraction	Sheet	A1 morbidity RR	A1 mortality RR	A2 MD lens (trials)
Published (human)	—	0.778 [0.567; 1.068]	1.021 [0.446; 2.337]	-0.24 [-0.32 ; -0.16] (7)
Layer 2 (source)	—	0.787 [0.580; 1.068]	1.023 [0.446; 2.346]	-0.24 [-0.32 ; -0.16] (7)
gemma4:12b	v1	0.779 [0.569; 1.065]	1.019 [0.445; 2.336]	-0.27 [-0.38 ; -0.17] (7)
	v2	0.779 [0.569; 1.065]	1.078 [0.470; 2.471]	-0.21 [-0.23 ; -0.19] (4)
qwen3:14b	v1	0.779 [0.569; 1.065]	1.021 [0.446; 2.337]	-0.42 [-0.91 ; 0.06] (7)
	v2	0.779 [0.569; 1.065]	1.019 [0.445; 2.336]	-0.22 [-0.59 ; 0.15] (3)
llama3.1:8b	v1	0.738 [0.510; 1.070]	1.023 [0.447; 2.344]	-0.63 [-1.03 ; -0.24] (5)
	v2	0.767 [0.552; 1.066]	1.023 [0.447; 2.344]	-0.58 [-0.85 ; -0.31] (5)
qwen3.5:9b	v1	0.779 [0.569; 1.065]	1.021 [0.446; 2.337]	-0.63 [-1.07 ; -0.19] (5)
	v2	0.779 [0.569; 1.065]	1.019 [0.445; 2.336]	-0.88 [-2.14 ; 0.39] (3)
deepseek-r1:14b	v1	0.788 [0.579; 1.072]	0.980 [0.428; 2.246]	-0.46 [-0.66 ; -0.25] (5)
	v2	0.769 [0.558; 1.061]	1.021 [0.446; 2.337]	-0.52 [-0.83 ; -0.21] (2)
qwen3.8:27b	v1	0.787 [0.580; 1.068]	0.400 [0.018; 9.121]‡	-0.25 [-0.37 ; -0.12] (7)
	v2	0.779 [0.569; 1.065]	1.023 [0.446; 2.346]	-0.20 [-0.34 ; -0.05] (4)

Table 11: The six-model campaign, Anchor 3: the two cells that decide (the methylene-blue arm of Aguilar 2016 and of Shaker 2025, as each sheet delivered them after the corrector’s frozen reduction; the review’s errata read 24/30 and 15/30, the sources 6/30 and 15/60), the 32-cell score, and the pooled odds ratio with its 95% confidence interval under the campaign engine (DerSimonian–Laird) and under the review’s own estimator (Paule–Mandel with Hartung–Knapp), with the trials pooled in parentheses; the first two rows are the review’s published cells and the source-verified cells under the same two engines. NR: the sheet returned no mortality entry for the trial. ‡as in Table 8.

Extraction	Sheet	Aguilar MB	Shaker MB	Cells	OR, DL (trials)	OR, PM–HK
Published (human)	—	24/30	15/30	29/32	0.737 [0.449; 1.208] (8)	0.73 [0.40; 1.36]
Layer 2 (source)	—	6/30	15/60	32/32	0.484 [0.319; 0.735] (8)	0.484 [0.342; 0.685]
gemma4:12b	v1	NR	15/60	23/32	0.519 [0.321; 0.839] (6)	0.520 [0.270; 1.002]
	v2	NR	15/60	23/32	0.418 [0.246; 0.711] (6)	0.418 [0.211; 0.828]
qwen3:14b	v1	6/30	15/60	32/32	0.484 [0.319; 0.735] (8)	0.484 [0.342; 0.685]
	v2	6/30	15/60	24/28‡	0.481 [0.311; 0.745] (6)	0.481 [0.393; 0.589]
llama3.1:8b	v1	6/30	15/60	26/32	0.496 [0.318; 0.773] (7)	0.496 [0.319; 0.770]
	v2	6/30	NR	23/32	0.517 [0.322; 0.832] (6)	0.517 [0.348; 0.770]
qwen3.5:9b	v1	6/30	15/60	28/32	0.460 [0.299; 0.708] (7)	0.460 [0.325; 0.651]
	v2	NR	15/60	16/28‡	0.450 [0.267; 0.758] (4)	0.450 [0.212; 0.951]
deepseek-r1:14b	v1	6/30	15/60	32/32	0.484 [0.319; 0.735] (8)	0.484 [0.342; 0.685]
	v2	6/30	15/60	22/24‡	0.548 [0.362; 0.831] (6)	0.548 [0.388; 0.775]
qwen3.8:27b	v1	NR	15/60	28/32	0.493 [0.315; 0.770] (7)	0.493 [0.326; 0.744]
	v2	NR	15/60	28/32	0.493 [0.315; 0.770] (7)	0.493 [0.326; 0.744]

Anchors 1 and 2 at the new context (H12.5). The first campaign’s pattern held: every model on both sheets reproduced Anchor 1’s morbidity pool within ± 0.01 (0.738–0.788 against 0.787), and the 27B build kept the top anchor-1 score on both sheets. **gemma4:12b**’s anchor-2 lens landed again at exactly -0.27 [-0.38 ; -0.17], inside its ± 0.05 band, but its anchor-1 v1 agreement rose to 109/124 (108 mechanical) against the band 103 ± 4 — one cell above it. By the pre-registered rule the context change from 16,384 to 24,576 tokens is a confounder for every comparison between the two campaigns, and it is declared here before any anchor-3

result is interpreted; the direction (upward, by one cell, on the descriptive anchor only) bounds its size.

The quotation sheet (H12.3) and the provenance nets (H12.4). Under v2 every model scored the same or lower on anchor 1 — mean change -6.5 cells of 124 (`qwen3:14b` -19 , `deepseek-r1:14b` -7 , `llama3.1:8b` -6 , `qwen3.5:9b` -4 , `gemma4:12b` -2 , `qwen3.8:27b` -1), against the pre-registered floor of -4 — and v2’s pool was closer to layer 2 than v1’s in four or more models on none of the three anchors. Both criteria of H12.3 fail: at the scale the first campaign asked for, the mandatory verbatim quotation lowers cell agreement on the descriptive fields, where the source prints table rows rather than sentences, and does not bring the estimates closer to layer 2. The provenance nets, run over every v2 cell as the declared exploratory arm (H12.4), flagged 19–39% of the eligible anchor-3 cells on the quote-exists net and 7–43% on the value-in-quote net — ranges no wider than on anchors 1 and 2 (10–57% and 7–45%): the expected excess on the table-dwelling anchor did not appear.

Phase-4 adjudication of the 633 non-matching cells. Phase 4 adjudicated every non-matching cell with its quotation (297 on anchor 1, 273 on anchor 2, 63 on anchor 3). Two hundred and eighty-eight are faithful to the source in another encoding or population layer — 122 population-layer choices, 78 case-mix summaries, 54 percent-for-count re-encodings, 16 standard deviations derived by the key’s own rule from the analysed n the model read rather than the randomized n the key holds, 8 unit re-encodings (HbA1c in mmol/mol), 3 partial re-encodings, 6 cells for which the primary itself prints two contradictory counts, and 1 “not reported” written in words; 175 are omissions; 149 are reading slips, concentrated on the continuous anchor (117 of 149: another visit’s change, a between-group difference written as an arm’s change, a dispersion of the wrong type, a factorial cell for a margin); and 21 are grader-side artifacts adjudicated in the models’ favor. Those 21 are three instrument findings. Ten are the Castro seal-pair collision of the first campaign, recurring on every sheet that transcribed the cell. One is a lens miss: a sheet wrote the displaced image it read with a Unicode minus, and the reversal lens’s patterns do not match it. Ten are a limit the campaign had not met before: where the key derives a cell (change = final – baseline) and the seal displaces one term, a model that applies the key’s own rule to the text it read lands on a number the lens has never seen ($5.3 - 6.9 = -1.6$ where the key holds -0.9), so the lens penalized a correct derivation; four models, both sheets, one trial. Because that trial contributes to Anchor 2’s pool, the lens pools of Table 10 carry the artifact: recomputed with those cells at the value the lens should have returned, `qwen3:14b` v1 moves from -0.42 to -0.32 , `llama3.1:8b` v1 from -0.63 to -0.49 , `deepseek-r1:14b` v1 from -0.46 to -0.27 and `qwen3.5:9b` v2 from -0.88 to -0.32 , a sensitivity recorded beside the primary that changes no hypothesis verdict. Thirty-nine cells put the key on trial and are listed for the author: three trials whose randomized and analysed counts are both printed, where the key holds the layer the review pooled and the models wrote the literal randomized figure. Sixteen cells carry a value the text does not print, the class the first campaign’s invention screen would flag: nine from `llama3.1:8b` on Anchor 1 — eight ratio-style ASA strings of the kind Supplementary Table S6 records and one count reconstructed with the other arm’s denominator — and seven on Anchor 2 from four models (three dispersions, three change means and one denominator), each a computed value with no reconstructible rule. None occurred on Anchor 3; all sixteen are extraction cells, and none reached an arithmetic answer.

4 Discussion

This study measured, in one continuous pre-registered chain, what small local models running on an integrated GPU can contribute to meta-analysis — and where the risk concentrates.

Under the perturbation control, none of these results is attributable to memory of the published review.

The division of difficulty appears structural as much as cognitive. Dichotomous material is transcription — counts and denominators printed redundantly across table, prose and percentage — and the pipeline *dampens* what error remains: a risk ratio is a ratio, so a different population layer shifts numerator and denominator together, and the weights derive from the counts themselves. Table 4 shows it: six models took visibly different denominator routes and still landed within a hundredth of the published morbidity pool. Each continuous quantity, by contrast, is printed once, in one of several competing layers, and the decisive step is semantic — identifying what a printed dispersion *is*. A mislabeled type sends the conversion down the wrong route, and inverse-variance pooling then *amplifies* the error, because weight scales with $1/SD^2$ [Higgins et al., 2023]: one dispersion four-fold too small dominated a pool and inflated its heterogeneity in the deployment phase. The same architecture is thus error-dampening for count outcomes and error-amplifying for continuous ones, which gives the promising-but-uneven extraction accuracy of the feasibility literature [Schmidt et al., 2024] a structure: in this record the unevenness tracks the outcome type at least as much as the model.

Nothing in the chain corrects a divergent cell: the downstream is deterministic, the nets only warn, the graders only classify, and the unperturbation lens is a pure table lookup. Table 5’s before-reversal column shows that neutrality mechanically: the four distant diamonds passed through the lens and stayed distant. The pools still match the published ones because the divergent cells of Table 2 concentrate in baseline descriptor fields the engines never consume, and in population-layer choices the ratio structure dampens. The six diamonds spread widely through reading-class differences alone: one engine computed all six. Because everything downstream of the sheets is code, the differences between models are differences in reading. The extension arm sharpens what *well-chosen* means: at a comparable footprint it carried roughly twice the parameters at about seven-tenths the precision (Section 2.6) and read better than every model in the cast on both anchors — and, run again as a peer in the second campaign, kept the top score on both descriptive anchors under both sheets (Table 8). On the count anchor the order changed: two 14B builds read all 32 cells and the 27B build 28, its four misses all omissions on the Spanish primary. So the parameter count tracked reading quality where reading is discrimination among printed layers, and not where it is transcription of counts — an observation, not a tested hypothesis, and a reminder that a cast’s ranking is anchor-dependent. Even the cast’s best diamond, **gemma4:12b**’s, falls short of the anchor’s homogeneity (Table 5) — the same effect, direction and significance over a noisier reconstructed evidence body, the residue plausibly injected by route-level differences (population layers, dispersion conversions) rather than by any single wrong cell — while the extension arm’s I^2 shows that homogeneity to be reachable by reading alone. “Beside the published value” is therefore a claim about the estimate, not digit-level equivalence of the whole meta-analysis; the wider interval is what a practitioner would inherit.

None of those reading failures was a fabrication: the invention screen found no cell whose numbers cannot be traced to the text read, and the inventory — omissions, population-layer choices, format variants, mislabeled quantities, replicate-stable in the stronger readers — is systematic and auditable, not stochastic. That profile is what makes a warn-only architecture viable: nets suffice for traceable errors, whereas fabricating models would demand the value-substituting interventions the doctrine forbids. A cloud-model extraction study reports the same shape — across 160 data items its errors were few and *missed data was the most common error type* [Gartlehner et al., 2024] — so omission dominating fabrication at opposite ends of the size and deployment spectrum suggests a property of the extraction task, not of one model class. The pooled-arithmetic ceiling agrees: even with the calculator, the list-valued pooling calls stayed beyond the whole 8–14.8B cast — which is why the architecture moves pooling into code rather than relying on the reader’s size. Hence the triage: dichotomous evidence tables

may be delegated broadly; continuous outcomes concentrate the model risk and deserve the strongest reader and the strongest checks. The second campaign’s 633 adjudicated cells give that triage its inventory: on the count anchor the non-matching cells are almost all omissions and the seven slips are legible in a sentence each; on the diet anchor 117 of 149 reading slips live — another visit’s change, a difference between groups written as an arm’s change, a dispersion of the wrong type — and there, too, sits the campaign’s residue of sixteen cells the text does not print: seven computed dispersions, change means and one denominator with no reconstructible rule, from four models, and nine from the first campaign’s weakest reader: eight ratio-style strings of the kind it had already produced and one count reconstructed with the wrong denominator.

The third anchor answers the question the first campaign could only pose. Its published diamond can be reproduced only by reproducing two transcription errors, and no sheet of any model did: the four models that read the Spanish primary read it as the source prints it, all six enumerated the three-arm trial, and two pooled to the corrected estimate to the third decimal — the significant benefit the review’s own Table 1 records and its forest plot denies. Three things scope that result, and the protocol wrote them down before the first call. The errata were raised by this study’s adjudicator during key construction, so the models did not *find* them; they were measured against them. The anchor-3 sheet asks four fields per arm over enumerated arms, which are the two axes on which the human extractors failed, so the instrument helps a reader land the correction. And the event counts carry no reading proof — the proof rests on displaced denominators in four and a half of eight trials — while the anchor’s primaries, unlike the two post-cutoff reviews, fall inside every cast model’s training window; a recited death count cannot be excluded on this anchor, though a recited *review* cell, the 24/30 that would have betrayed recall of the meta-analysis rather than reading of the trial, never appeared. What the anchor does establish is narrower and useful: a source-faithful local reader, graded against the primary rather than the review, produces the corrected meta-analysis without any instruction to look for errors, and a benchmark that graded against the published cells would have scored its two best readers zero on the two cells that matter.

Three design choices in that anchor are the author’s to disagree with, and each is stated here so that disagreeing requires no reconstruction. First, the reading proof: the alternative to displacing the denominators with coherent recomputation was to displace the percentage alone, which makes count, denominator and percentage contradict one another inside the source — a defect a careful reader can notice and repair, contaminating the very measurement the proof protects — or to declare the anchor unperturbed and lose the proof entirely; the chosen design buys a partial proof at the cost of a new operator capability, and its limit (counts unproved, three trials uncovered) is printed beside every anchor-3 number. Second, the target analysis: the review describes its primary analysis three incompatible ways, and the forest plot governs here because it is the only description with cells attached; the other two are reported, not reconciled, and a reader who prefers the text’s composition inherits a different set of trials and no cells to grade. Third, the outcome window: Aguilar’s 21-day figure is pooled into an analysis the review labels 28–30 days, because that is what the review did, and its exclusion runs beside it as a registered sensitivity — 0.484 with the trial, 0.493 without, the conclusion unchanged either way, which was known before the run rather than discovered after it.

The mandatory-quotation sheet was the first campaign’s open question, and the second closes it in the negative at the scale it asked for. The quotation costs accuracy where the source prints table rows rather than sentences, most for the reader that follows the instruction most literally, and buys nothing at the estimate. Its value, measured in the first campaign’s editorial reading, is elsewhere: a quotation lets a reviewer adjudicate a cell without opening the article. Whether that is worth 6.5 cells in 124 is a deployment decision, not a benchmark result, and the sheet is not adopted as the default here. Two smaller results belong beside it. The context change that Anchor 3 forced moved one pre-registered control by one cell, upward, on

the descriptive anchor only, and is declared a confounder for every cross-campaign comparison; its size bounds what it can explain. And the provenance nets, expected to over-fire on an anchor whose cells live in table rows, fired no more than on the prose-and-table anchors — so the table-cell blind spot the first campaign measured is a blind spot of the *quotation*, not of the nets.

The deployment phase turns the benchmark toward real use, where no answer key exists. At all seven cells of the frozen discrepancy panel the model sided with the *source* against the published value (the other four of the eleven fall outside it: a method-caption error, a sample-size convention spanning six trials, and two cells whose sources report nothing verifiable to side with; Table 1). A local reader plus a three-way table makes such discrepancies *visible* instead of requiring trust in either extractor. One carries conclusion-level weight (Table 1, #16): the anchor’s significant ileus reduction (abstract RR 0.48) rests on the mislabeled pulmonary-complication row, and over the two genuinely-ileus trials the pool crosses 1 under either method; its no-difference conclusions for morbidity and mortality survive every documented discrepancy, as the six-model reproductions demonstrate. This is consistent with reports that model assistance can match human extraction quality in live reviews [Lee et al., 2026], and adds that disagreement informs in both directions: the same source-verification process corrected this study’s own key twice during development.

The phase’s one failed tolerance decomposed to a single cell, where the source prints the same quantity twice — a table interval and a prose “ $\pm$ ” — and the model transcribed one layer under the wrong label. That ambiguity in the source’s own notation seeds the defect, which defines this architecture’s residual failure class: a model that misdeclares its sheet escapes every check that trusts the sheet. The class is not peculiar to machine readers. An audit of a published sports-medicine meta-analysis found the same dispersion mix-up in 9 of its 16 effect sizes; correcting it moved the pooled estimate from a standardized mean difference of 1.46 [0.91, 2.01] to 0.54 [0.37, 0.71] and cost a second comparison its significance, over a trial literature in which nearly one report in five describes variability with a standard error rather than a standard deviation [Sandercock, 2024]. Human extractors disagree on exactly these quantities: in an observer-agreement study of continuous outcomes, independent extractions of the same trials yielded meta-analyses that disagreed [Tendal et al., 2009]. The class therefore deserves a dedicated sheet-versus-source dispersion net rather than a better model, in any pipeline, human or machine. The instrument asymmetry likely mattered: the continuous sheet carries no provenance field, whereas the dichotomous sheet’s mandatory **where** locator repeatedly disambiguated population layers during adjudication; extending provenance to every sheet is the corresponding instrument work, never retroactive.

Detection layers are complements, not substitutes. Conversational nets catch conversation-layer defects but are structurally blind to a well-formed poisoned cell, which only the sheet-versus-source net saw; the weight-share product flag was blind too, because random-effects weighting re-equalizes shares. P4 supplies the complementary case: confirm-or-correct resolved every warning that fired — the net states what it saw, the model confirms or revises, no value is ever overwritten — after which the model’s own pooling call closed digit-consistent with its executed results. A net that can only point still changes outcomes, provided a model is there to answer it. No warn-only configuration made the wrong number right — by doctrine it must not — so the safest key-free architecture this record supports is a deterministic engine, sheet-versus-source nets, a conversational harness for orchestration slips, and a reviewer dispatched to the primary’s table whenever any flag fires.

Unlike most of the literature it joins, everything measured here runs locally, at zero marginal cost, on hardware that keeps clinical text on-premises; the perturbation proof, the public seals and the adjudication rite operationalize the conditions the evidence-synthesis organizations attach to such use [Flemyng et al., 2025]. That reading-versus-recall control matters more, not less, as anchors and primaries enter training corpora [Xu et al., 2024], and the two operator gaps

behind this campaign’s five mechanical candidates are now demonstrated across architectures and queued as instrument fixes.

These discrepancies also measure the human side of extraction, and the two extractor kinds fail differently — with one telling overlap. That the human column carries errors at all is consistent with the methodological literature rather than remarkable within it [Gøtzsche et al., 2007, Jones et al., 2005, Tendal et al., 2011, Mathes et al., 2017]. What this study adds is the cell-by-cell source-quotation trail and a model column audited beside the human one — on one ruler: scored against the source-verified layer exactly as the models are, the human extraction of Anchor 1 agrees in 102 of 124 cells, that of Anchor 2 in 49 of 49, that of Anchor 3 in 29 of 32 (Tables 2 and 8), so on the first anchor the published review sits below the cast’s better readers and above its weakest, and on the third below its two best. The anchor’s eleven source-confirmed discrepancies (Table 1) are transcription and labeling slips, and the human extractors fabricated almost nothing — two cells of hours the source never prints (#9) are the exception; the model profile concentrates instead in omissions, layer choices and format variants. The overlap is the instructive part: the one human error with conclusion-level weight and the one model error that distorted a pool are the *same class*, a mislabel — an outcome heading, a dispersion type — suggesting that label semantics, not numbers, is where extraction fails for both kinds of extractor, and therefore where verification effort should concentrate.

Plausibility bias — any judge’s tendency, human, hybrid or model, to prefer a plausible reading over the printed one — runs through the instruments and the findings alike. It shaped the instruments before a single campaign run: both the adjudication rite and the warn-only doctrine exist because of it (Sections 2.4 and 2.10). In the campaign the bias was measured, not presumed, and appeared exactly once — on the judge’s side: the deployment record initially charged the model with *computing* the mislabeled dispersion, until a detection-net run showed the value printed verbatim in the source’s own prose and the charge was corrected with a dated note in the archived record [Barra, 2026]. The same containment caught the grading layer again, mechanically rather than judgmentally, in the four verbatim-correct transcriptions that a seal-pair lookup collision had charged as divergences and that were adjudicated in the models’ favor (Supplementary Table S7) — and three more times in the second campaign, where 21 of 633 charges were the grader’s: the same collision on every sheet that transcribed the cell, a reversal lens that does not recognize a Unicode minus, and, most instructive, a lens that cannot reverse a quantity the model correctly *derived* from the displaced text, so that the four models which applied the key’s own rule to what they read were charged for it. The model stages showed zero occurrences: no value was ever substituted (structurally forbidden), the sign-echo net never fired, and all six transcribed even the sign-sensitive cells verbatim — a factuality profile opposite to the judges’. This aligns with documented model-judge biases [Zheng et al., 2023], which appear to generalize beyond model judges: the constraint that works is structural — quote before verdict, flag before fix — not a better judge. The same constraint addresses automation bias, the documented tendency of users to accept a system’s output in clinical decision support [Goddard et al., 2012]: a net that only points leaves the value, and the decision, to the reader.

The instruction-language robustness check closes a confounder rather than opening a finding. A referee could ask whether drafting the instruments in Portuguese and freezing them in English handicapped — or peculiarly helped — models trained predominantly on English; per phase the two records agree within replicate-level variance (Table 7). This is consistent with adequate multilingual instruction-following at these model sizes for structured tasks — a claim scoped to these instruments — and establishes that no conclusion above is an artifact of instruction language.

Several limitations bound these claims. Three anchors and 29 primaries are the whole evidence base — two anchors from one journal family, the third from another publisher and domain; every superlative is scoped to these corpora, and no formal between-model inference is

attempted. One quantization per model and one hardware profile were tested, so precision and parameter count are not separable here: the sixth model entered the first campaign after the fact and was not taken through P4, and although the second campaign ran it as a peer, its advantage is still measured against five builds that differ from it in both. The second campaign’s context differs from the first’s by 8,192 tokens, and one of its two controls fell one cell outside its band, so cross-campaign comparisons carry a declared confounder. On Anchor 3 the reading proof covers the denominators of four and a half trials and no event count, and the primaries predate the models’ training cutoffs; that anchor’s errata were raised by the adjudicator before any model read it, and its sheet is built on the axes where the human extractors failed. The adjudication of the second campaign’s 633 cells was delegated by the author to the assistant, under the rite and with every verdict reversible, which trades independence for completeness and says so. The human rows of Tables 2 and 8 score the published cells against a key that was built by verifying those same cells, so their divergences are, by construction, the verification record itself; the rows place the reviews on the models’ ruler and support no inference beyond that. The mechanical cell comparator is a declared approximation of the full adjudicated ruler, so absolute cell scores understate agreement and cross-model gaps partly reflect format habits. The perturbation operator’s residual gaps (derivable totals; prose restatements) are documented and exploitable, though they produce reconstruction, not recall. The English harness build reproduces the frozen net logic but was exercised in one formal run of one model. The adjudicator is the same human–AI pair that built the harness, mitigated by mandatory quotation and public self-correction rather than independence, and the known biases of model judges [Zheng et al., 2023] are constrained, not eliminated, by that rite. Finally, the deployment phase’s clean texts cannot separate reading from recall by design; its claims rest on the perturbed phases’ proof.

In conclusion, a consumer machine running one well-chosen local reader — a 12-to-14-billion-parameter model at 4.5 bits per weight, or, at the same per-call latency and a comparable footprint, 27 billion parameters at 3.13 — a frozen deterministic engine and warn-only detection nets can rebuild published meta-analyses to the published digits, expose documented errors in them with source quotations, and fail in ways that are named, single-celled and flaggable — and none of this depends on the instruction language. The generalization gate the first campaign designed has now been run: on a third anchor, in a third domain and a third estimand, the same frozen architecture rebuilt the meta-analysis as the sources support it rather than as published, and the two differ in their conclusion. The next question is how often grading against the primaries, rather than against the review, would change published estimates if applied at scale.

Data and code availability

All protocols (pre-registered, with dated amendments committed before each run), the frozen English instrument library with its Portuguese-correspondence tables, perturbation seals with SHA-256 fingerprints, the two-layer answer keys and the anchor source-verification record, the deterministic engines and graders, the English harness build, all raw model outputs and run logs (277 from phases P1–P4, 54 from the extension arm, the deployment phase’s 42 extraction calls and harness runs, and the second campaign’s 696 sheets with their per-call records), per-phase evaluation records, the second campaign’s protocol with its pre-registration review, its Anchor-3 key, sealed perturbation map, errata record and held alert table, and its Phase-2 and Phase-4 records with the 633 adjudicated cells, adjudications with source quotations — consolidated in a raw-data index of the adjudication layer (`dados/adjudication-record.md`: the quotation-bearing key cells, the judge’s reversals and withdrawn accusations, the campaign-era correction with its refuting net run, and every flag resolution) — and the exact quantized builds (pinned by Ollama digest) are public at <https://github.com/gbbarra/llm-evidence-extraction-benchmark-ptbr> and archived at Zenodo [Barra, 2026]. Closed-stratum primary articles are not redistributed; scripts rebuild that corpus stratum from legally obtained copies.

Declarations

Funding. This work received no external funding. All computation ran on the author’s own consumer hardware; neither the author nor any institution received payments or services from a third party that could be perceived to influence this work.

Competing interests. The author declares no competing interests. For transparency: the AI assistant declared below (Claude, Anthropic) was used under an ordinary consumer subscription paid by the author, and the author has no financial or professional relationship with Anthropic or with the organizations whose open-weight models are evaluated here.

Use of artificial intelligence. This campaign was designed and conducted by the author with the assistance of Claude (Anthropic), which translated and froze the English instruments, implemented the phase runners, the English harness build and the graders (all committed before their runs), performed the mechanical adjudications under the quotation rite, drafted the second campaign’s protocol from the author’s six Phase-0 decisions (with the three assistant-originated design choices flagged for the author’s ruling before any run), built its Anchor-3 key, perturbation instrument and harness, performed the second campaign’s Phase-4 adjudication of 633 cells under the author’s delegation with every verdict quotation-bound and reversible, and drafted this manuscript under the author’s direction and review. The evaluated local models produced only the extraction sheets, the arm-A/B arithmetic answers and the phase-P4 harness calls graded here; every downstream number in this article was produced by the frozen code and verified mechanically. Per the position of the evidence-synthesis organizations [Flemyng et al., 2025], every AI contribution is reported and the author retains responsibility for the work.

Ethics approval and consent. Not applicable. This study involved no human participants and no new data collection from humans: it analyzes only published, aggregate-level reports of randomized controlled trials and two published meta-analyses. No institutional review board approval or participant consent was required.

Acknowledgments

I thank my family, for the support that made this work possible; my teachers and professors, whose lessons are present throughout it; and Sabin Diagnóstico e Saúde, where I work, for the daily practice of diagnostic medicine in which the habits of measurement behind this project were formed.

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Supplementary material: P1’s 124 comparator failures across the five models, cell by cell

This supplement enumerates every non-matching cell of the READ phase (Table 2): all 124 cells, across the five models, where a sheet value — after seal reversal — failed the numeric comparator against the source-verified key. Each cell was re-examined against the perturbed source text its model actually read, and classified by mechanism; the class totals are exhaustive ($12+27+26+34+12+9+4 = 124$). Model abbreviations: G = `gemma4:12b`, Q14 = `qwen3:14b`, L = `llama3.1:8b`, Q35 = `qwen3.5:9b`, D = `deepseek-r1:14b`; a parenthesized “(2)” after a field means both arms’ cells. The reading of the whole: no cell is an invention — every value traces to the text read; 87 cells (Tables S2–S4) are faithful to the source in a different encoding, population layer or resolution; 12 cells (Table S5) are genuine reading slips — a real value taken from the wrong row, arm or scope; 9 cells (Table S6) are ratio-style re-expressions matching no printed layer, all from the cast’s weakest reader; and 4 cells (Table S7) were adjudicated in the models’ favor as a grader-side artifact. The extension arm’s 17 divergent cells (Section 3.1) are enumerated, under the same classes, in the repository’s Amendment-1 record (`dados/estudo8/avaliacao-amend1.md`). The second campaign’s 633 non-matching cells (Section 3.7) are enumerated cell by cell, grouped by trial, field and written value with the deciding quotation and the rule or reader that decided each, in the repository’s Phase-4 record (`dados/estudo12/p4/adjudicacao-p4.md`); its class totals are: 122 population-layer choices, 78 case-mix summaries, 54 re-encodings, 3 partial re-encodings, 16 dispersions derived with the model’s own n, 8 unit re-encodings, 6 cells for which the primary prints two contradictory counts and 1 “not reported” written in words (288 faithful); 167 omissions and 8 label-only cells (175); 95 row or scope slips, 28 wrong-type dispersions, 16 unmatched values, 6 arm swaps, 2 polarity inversions and 2 field mixes (149); and 21 grader-side artifacts (10 lookup collisions, 1 lens miss, 10 quantities derived through the seal).

Table S1: Omissions (12 cells): the sheet says *not reported* where the source prints a value. The Diaper rows carry a nuance the class name hides: the source prints only a partial datum (the ASA III–IV share), and writing NR for a full-distribution field is defensible conservatism; the key transcribes the partial datum.

Trial	Field	Source value (the key)	NR written by
Sun	inotrope use	GDFT 18 (36%) vs. control 27 (54%)	Q14, Q35
Wu	inotrope use	norepinephrine 15 (25.9%) vs. 24 (42.9%); phenylephrine 12 (20.7%) vs. 24 (42.9%); ephedrine 12 (24.0%) vs. 19 (35.2%)	Q14, Q35
Redondo	ASA (2)	0:10:6:0 (control); 0:13:6:0 (GDFT)	Q14, D
Calvo			
Diaper	ASA (2)	only the ASA III–IV share is printed: GDFT 98 (50.0%) vs. control 85 (42.9%)	Q14, Q35

Table S2: Equivalent re-encodings (27 cells): the same printed datum re-expressed in another notation; in every row the arithmetic was verified against the key’s counts (e.g. Castro control ASA II: $26/42 = 61.9\% \approx 62$). The comparator, which matches digits, cannot see these equivalences — the limitation Table 2’s caption declares.

Trial	Field	Key (source layer)	What was written
Castro	ASA (2)	11:26:5:0 / 7:31:5:0	counts as percentages, 26/62/12 and 16/72/12 — as % strings (G), as a ratio string (L); all match the counts over the arm totals
de Waal	ASA (2)	24:123:86:1 / 17:132:95:4	as percentages 10.3/52.6/36.7/0.4 and 6.9/53.2/38.3/1.6 (Q35, D); exact over the analysis-set totals
FEDORA	ASA (2)	35:136:40:0 / 14:122:73:0	as percentages 16.6/64.4/19.0 and 6.7/58.4/34.9 (D); exact
Wu	ASA	0:22:34:0 (control); 0:25:33:0 (GDFT)	control as percentages 39.3/60.7 (L, D); GDFT as percentages 43.1/56.9 (D); exact
FEDORA	laparoscopy (2)	109 / 105	as percentages 51.7% (= 109/211) and 50.2% (= 105/209) (D)
Yoon	laparoscopy (2)	0 (open surgery)	“0%” (D) — same meaning
Sujatha	laparoscopy (2)	0 (open surgery)	“excluded” (G) — same meaning, no digits for the comparator
Arslan-Carlon	postoperative ileus (2)	30 (21.3%) / 36 (25.4%)	rearranged with denominator and rounded: “21% (30/141)”, “25% (36/142)” (G, L, Q35, D) — counts identical after seal reversal; the raw sheets show the perturbed image (31) that reversal maps back to 36, i.e. faithful perturbed-world reading

Table S3: Population-layer choices (26 cells): the model filled an *n randomized* cell with a different participant count printed by the same trial. The key anchors each cell on the source-verified layer of the anchor’s own cell, so the direction varies by trial — for de Waal the key holds the randomized counts and models wrote analysis-population numbers; for FEDORA and Hokenek the key (like the anchor) carries the analysis set and models wrote the allocation, the field name’s literal layer. The dampening mechanism discussed with Table 4 is why none of this moved a pooled estimate: ratio outcomes shift numerator and denominator together.

Trial	Field	Key (source layer)	What was written
de Waal	n randomized (2)	274 / 259 (randomized)	printed analysis populations: 258/244, the complications analysis (“PGDT group (n = 258)” ... “control group (n = 244)”) (G, L); 248/234, the hypotension analysis denominators (Q14); and mixtures reusing the control arm’s 259 (Q35, D)
FEDORA	n randomized (2)	211 (control) / 209 (GDFT), the analysis set	226 (control) / 224 (GDFT) — the allocation: “450 patients were randomized to the GDHT group (n=224) or control group (n=226)” (G, Q14, L, Q35)
Hokenek	n randomized (2)	39 / 39 (analysis set)	40 / 40 — the allocation: “(control group: 40, PVI group: 40)” (G, Q14, L)
Sujatha	n randomized, control	102 randomized	101 — the analysed count, the layer the anchor’s own cell carries (L)
Diaper	n randomized, control	201	200 — the other arm’s allocation; the power calculation also prints “200 patients in each group” (L)

Table S4: Case-mix summaries (34 cells): the *surgery type* field condensed to a category label where the key transcribes the source’s enumerated case mix. Faithful but lossy — no wrong fact, less resolution than the key’s reading.

Trials	Models	Example
Yoon, Sun, Sujatha, Diaper, de Waal, all five FEDORA		Yoon — key: “Eight open radical cystectomies with ileal conduit, 31 open...”; written: “open radical cystectomy”
Schmid	G, Q14, L, Q35	key: “16 Whipple, 71 oesophageal, three pancreatectomies...”; written: “major non-cardiac surgery” (G), “major abdominal surgery” (L)

Table S5: Different row or scope picked (12 cells): genuine reading slips — the value exists in the source, in the wrong place. These, not fabrications, are the READ phase’s real failure inventory.

Trial	Field	Key (source layer)	What was written
Castro	blood loss (2)	the “Intraoperative bleed-ing” row: 1283.2 ± 959.7 / 1100.1 ± 851.1	the same table’s “Losses” row: 5062.9 ± 3287.5 / 4364.5 ± 2920.6 (G)
Castro	colloid (2)	Voluven (mL): 737.8 ± 383.2 / 779.7 ± 565.1	the adjacent per-kilogram row, Voluven (mL kg^{-1}): 10.7 ± 5.9 / 10.8 ± 6.6 (L)
Sun	ASA (2)	8:33:9:0 / 6:36:8:0	adjacent rows mixed: control “16 (32%)” is the table’s Female row; the GDFT ASA-I cell carries the control arm’s 8 (D)
Wu	ASA, GDFT	0:25:33:0 (= 43.1/56.9%)	the two shares transposed: 56.9/43.1 (L)
FEDORA	morbidity, control	35 (16.6%)	36 — a nearby printed value (L)
Schmid	ASA, GDFT	NR (source prints eligibility criteria only)	a distribution filled from another scope (L, D)
Arslan-Carlon	ASA, GDFT	NR (no per-arm distribution in the text)	the overall “30% II / 70% III–IV” placed in the per-arm field (D); placeholder junk “I:II:III:IV (NR:NR:NR:NR)” (L)

Table S6: Unmatched re-expressions (9 cells, all **11ama3.1:8b**): ratio-style strings that match no printed layer as a set. The digits are individually present in or derivable from the text read — which is why the invention screen does not fire — but the assembled distributions correspond to nothing the source prints; sheet residue, held for adjudication. The same model is also the cast’s least replicate-stable (17/36 of its divergents stable), so part of this residue may be sampling noise rather than a systematic reading.

Trial (ASA field)	Key (source layer)	11ama3.1:8b wrote
Yoon (2)	2:27:7:0 / 4:29:6:0	0.75:0.24:0.01 / 0.74:0.25:0.01
de Waal (2)	24:123:86:1 / 17:132:95:4	1:2.9:3.2:0.6 / 1:2.8:3.2:1.2
Diaper (2)	only the III–IV share is printed	full I–IV ratios 47:38:22:13 / 50:50:30:10
Redondo Calvo (2)	0:10:6:0 / 0:13:6:0	class-shifted 10:4:2:0 / 13:3:0:0
FEDORA, GDFT	14:122:73:0	partial shift: I:122, II:148, III:73, IV:0

Table S7: Grader-side lookup collision (4 cells): the models were right. Adjudicated with the source open, in the same quotation-bound rite the key itself is subject to; Table 2 reports the adjudicated scores, and the comparator’s raw output is preserved in the archived record. The same mechanics that caught the human anchor’s discrepancies caught the grader here.

Trial	Field	What happened
Castro	blood loss, control	Q14, L, Q35 and D transcribed the printed “Intraoperative bleeding (mL) 1283.2 ± 959.7 ” verbatim — byte-identical to the key’s source value. The unperturbation lookup then applied the trial’s seal pair $31 \rightarrow 28$ (an ASA count) as a substring replacement <i>inside</i> “ <u>1283.2</u> ”, yielding “1313.2” and failing the comparison. Correct transcription; comparator artifact. (G alone genuinely diverged on this cell — the “Losses” row; Table S5.)